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2025 Stock Assessment of Striped Marlin in the Southwest Pacific Ocean: Part II Data-moderate approach

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1 Executive summary

2 Introduction

Assessments of Southwest Pacific Ocean (SWPO) striped marlin (*Kajikia audax*) have been challenging. As documented in the joint NOAA-SPC modeling meeting report (Ducharme-Barth et al., 2025), key issues identified by WCPFC SC20 included poor fits to size composition data and relative abundance indices, conflicts between different data sources, and difficulties in estimating model initial conditions.

Discussions at the 2025 SPC Pre-Assessment Workshop highlighted additional challenges related to the estimation of total population scale. The low biomass scale indicated by previous versions of the assessment appears largely driven by the size composition from multiple fisheries. While estimation of low biomass scale is not inherently problematic, the resulting high ratio of fishing mortality (F) to natural mortality (M) that is maintained over multiple decades in the face of relative stability in catches and catch-per-unit-effort (CPUE) was noted as suspicious and flagged for further investigation.

The complexity of integrated age-structured models, while providing detailed population dynamics representation, can become problematic when fundamental data conflicts exist or when key biological parameters are uncertain. For SWPO striped marlin, these challenges are compounded by spatiotemporal heterogeneity in fleet coverage, potential non-representativeness in mixed-fleet composition data, and limitations in age data from opportunistic sampling. Changes to productivity assumptions (growth, natural mortality, maturity, and/or steepness) or selectivity patterns will impact biomass scaling through predictions of expected size composition data.

Scaling back model complexity to a data-moderate approach, like Bayesian surplus production models (BSPMs), offers analysts a simplified yet robust alternative for stock assessment when data limitations or conflicts present challenges for more complex models. These models have proven effective in recent pelagic fish stock assessments (Winker et al., 2018), and are routinely applied for WCPFC shark assessments (ISC, 2024; Neubauer et al., 2019, 2024). They facilitate a more tractable exploration of model assumptions by distilling productivity and fishing assumptions into a restricted parameter subset. The BSPM framework allows for explicit incorporation of parameter uncertainty through informative priors while maintaining computational efficiency and interpretability.

One advantage of the Bayesian approach is the use of prior information to propagate uncertainty in model assumptions. Biological simulation approaches can provide robust methods for parameterizing key population dynamics parameters (ISC, 2024; Pardo et al., 2016). For species with complex life histories like striped marlin, these simulation-based priors can incorporate uncertainty in growth, maturity, natural mortality, and reproductive parameters to derive realistic distributions for maximum intrinsic rate of increase and carrying capacity. Additionally, prior pushforward checks (Kim & Neubauer, 2025; Monnahan, 2024) can be used to further refine parameter distributions by identifying parameter combinations that produce clearly improbable outcomes.

This analysis presents a data-moderate approach for SWPO striped marlin that addresses some of the key technical challenges identified in previous assessments while providing a robust framework for stock status evaluation. The model incorporates biological uncertainty through simulation-based priors and includes approaches to address catch uncertainty and population depletion. It is important to note that this analysis complements the 2025 revised stock assessment report (Castillo-Jordan et al., 2025). While the relative simplicity of data-moderate approaches offers some advantages relative to more data-intensive approaches like fully-integrated age-structured stock assessment models, particularly when uncertainty in data or key assumptions is high, this comes at the cost of simplifying the model fishery and population dynamics. Over-simplification of key age-structured processes, or lack of spatiotemporal specificity could result in biases in data-moderate approaches. Taking multiple modeling approaches into account can give managers a holistic view on the likely status and trajectory of the stock.

3 Methods

3.1 Model Framework

To address some of the issues raised in Section 2, a series of Bayesian state-space surplus production models (BSPMs) spanning the period 1952-2022 were developed following the Fletcher-Schaefer production model framework (Edwards, 2024; Winker et al., 2020). Development of the BSPM followed the approach of (ISC, 2024; Neubauer et al., 2019) and incorporated recent best practices for surplus production models (Kokkalis et al., 2024) and Bayesian workflows for stock assessment (Monnahan, 2024) as guides for model development, analysis, and presentation.

The models were implemented in the Stan probabilistic programming language (Team, 2024b) to take advantage of enhanced convergence diagnostics, greater efficiency in posterior sampling through the no-U-turn (NUTS) Hamiltonian Monte Carlo (HMC) algorithm (Betancourt & Girolami, 2013), and greater flexibility with model configuration and prior specification. Implementation in R using the *rstan* package (Team, 2024a) provides connection to an ecosystem of R packages for model diagnostics and validation: *bayesplot* (Gabry & Mahr, 2024) for posterior visualization and *loo* (Vehtari et al., 2024).

In their baseline configuration, the models began from an assumed unfished state in 1952 and incorporate process error in population dynamics, observation error in the relative abundance index and catch time series, and uncertainty in biological parameters. Population scale or carrying capacity $log(K)$ is given in the units of the catch which is total numbers. The models fit to a single time series of relative abundance (e.g., standardized CPUE index) at a time. Rather than subtracting the catch directly, the fishing mortality necessary to produce the catch was estimated internally by fitting to a time series of observed catches. Computationally, this was necessary to avoid placing hard constraints on the posterior parameter space which can cause convergence issues for the HMC sampling algorithm. Additionally, rather than directly estimating the independent fishing mortality values required to produce the catch, fishing mortality was derived from an estimated time-varying catchability trend and a time series of observed longline fishing effort. The primary rationale for this modelling choice being that it is easier to develop a defensible prior for catchability rather than an arbitrary prior for fishing mortality, given that previous work (ISC, 2024) showed a strong correlation between estimated population scale and the fishing mortality prior. Unless otherwise mentioned, all models used the same prior parameterizations (Table 2) based on the biological simulation filtering described in Section 3.2.

Using the baseline model as a starting point, a number of one-off sensitivity analyses were conducted to explore key model assumptions or data questions. These included: the choice of relative abundance index used to drive the model dynamics, how catch observation error was dealt with, sensitivity to assumptions in data errors for early model catch and effort observations, alternative priors for catchability, population carrying capacity, and production function shape parameter. Based on these results a structural uncertainty grid was developed for the provision of stock status and management reference points.

3.1.1 Input data

The BSPMs were fitted to catch and standardized CPUE data for SWPO striped marlin. Effort was used as an input to drive the time series of estimated fishing mortality via the estimated catchability trend.

Annual catch data (in numbers of individuals) spanning 1952-2022 were compiled from the 2024 assessment (Castillo-Jordan et al., 2024) sources and aggregated into total removals by year. The catch series (Figure 1) shows initially low removals in 1952-1953, followed by a high peak in 1954. Overall, however, catches have been relatively stable though catches in recent decades have shown a slight decline.

Annual longline effort (in numbers of hooks) matching the SWPO assessment domain, as specified in Castillo-Jordan et al. (2024), was extracted from publicly available WCPFC data. Longline effort in the assessment domain (Figure 1) increased gradually from zero in the early 1950s to ~100 million hooks per year in the early 1970s where it stabilized between 100-150 million hooks per year through the 1990s. Beginning in 2000 effort increased substantially to ~350 million hooks per year in 2003, and has fluctuated around that level for the last 20 years.

A number of indices were investigated either as one-off sensitivities or included in the structural uncertainty grid (Figure 2). These included the composite distant water fishing nation (DWFN) longline standardized CPUE index (1988-2022) from the diagnostic case of the 2024 assessment (Castillo-Jordan et al., 2024), an Australian longline index, a New Zealand recreational sportfish index, and three different observer data based indices (Neubauer et al., 2025). The Pacific Islands observer program indices contained information from New Caledonia, Fiji, Tonga, and French Polynesia observer programs. Three alternative indices were constructed either by including data from all programs together, including data from just the western programs (New Caledonia, Fiji, and Tonga) or including data just from French Polynesia. The DWFN, Australia and observer without French Polynesia indices show recent increases; the New Zealand and observer with only French Polynesia indices show recent declines; and the all observer program index shows no trend.

3.1.2 Population Dynamics

The population dynamics follow a hybrid Fletcher-Schaefer surplus production model (Edwards, 2024; ISC, 2024) with state-space formulation where population depletion x_t (relative to carrying capacity K) evolves according to:

$$x_t = \left(x_{t-1} + \text{surplus production}_{t-1}\right) \times \epsilon_t \times \exp(-F_{t-1})$$

where surplus production is defined as:

$$\text{surplus production}_{t-1} = \begin{cases} R_{Max} \cdot x_{t-1}\left(1 - \frac{x_{t-1}}{h}\right), & x_{t-1} \leq D_{MSY} \\ \gamma \cdot m \cdot x_{t-1}\left(1 - x_{t-1}^{n-1}\right), & x_{t-1} > D_{MSY} \end{cases}$$

Here, R_{Max} is the maximum intrinsic rate of increase, n is the shape parameter, F_{t-1} is the fishing mortality in year $t-1$, and ϵ_t represents multiplicative process error with $\epsilon_t = \exp(\delta_t - \sigma_P^2/2)$ and $\delta_t \sim N(0, \sigma_P)$. The intermediate parameters are: $D_{MSY} = (1/n)^{1/(n-1)}$, $h = 2D_{MSY}$, $m = R_{Max} \cdot h/4$, and $\gamma = n^{n/(n-1)}/(n-1)$.

Fishing mortality F_t is in turn defined as a function of fishing effort and time-varying catchability:

$$F_t = q_{eff} \times \exp(\delta_{q,t} - \sigma^2_{qdev}/2) \times \mathrm{E}_t$$

where q_{eff} is the base catchability coefficient, E_t is the fishing effort[1] in year t, and $\delta_{q,t}$ represents temporal variation in catchability effectiveness.

The catchability deviations $\delta_{q,t}$ follow a structured temporal process organized into discrete periods of length n_{step} years. In the baseline model $n_{step} = 1$ which allows for annual catchability deviates across 70[2] periods p. Catchability remains constant within periods p, and evolves between periods according to an AR(1) process:

$$\delta_{q,p} = \rho \times \delta_{q,p-1} + \epsilon_{q,p} \times \sigma_{qdev} \times \sqrt{1-\rho^2}$$

where ρ is the autocorrelation parameter, σ_{qdev} is the standard deviation of catchability innovations, and $\epsilon_{q,p} \sim N(0,1)$. The initial period is constrained such that $\delta_{q,1} = 0$. The term $\sqrt{1-\rho^2}$ normalizes the variance of $\delta_{q,p}$ to remain stationary at σ^2_{qdev} regardless of the value of ρ.

For each year t, the corresponding period assignment is:

$$\mathrm{period}(t) = \min\left(\left\lfloor \frac{t-1}{n_{step}} \right\rfloor + 1, n_{periods}\right)$$

and the catchability deviation for year t is assigned as $\delta_{q,t} = \delta_{q,\mathrm{period}(t)}$. This fishing mortality parametrization is a simplification of the *catch-errors* formulation in MULTIFAN-CL (Fournier et al., 1998) and allows catchability to vary smoothly over time while maintaining temporal structure and avoiding overfitting to annual fluctuations in the effort-catch relationship.

Model initial conditions are defined for $t = 1$ as $x_t = x_0 \times \epsilon_t$ where x_0 is the estimated depletion relative to unfished at the start of the model period. The prior specified for x_0 in the baseline case assumes an unfished initial condition with some uncertainty (Table 2).

3.1.3 Observation Model

The model fits to a standardized CPUE index using a lognormal likelihood:

$$I_t \sim \mathrm{Lognormal}(\log(q \times x_t) - \frac{\sigma^2_{O,t}}{2}, \sigma_{O,t})$$

where catchability q is analytically derived assuming an uninformative uniform prior, and total observation error $\sigma_{O,t} = \sigma_{O,input} + \sigma_{O,add}$ combines fixed input uncertainty with an estimated additional error component.

[1] When input in the model the raw input effort, defined as *hundred hooks* in the WCPFC database is divided by `1e6` to re-scale catchability so that it is numerically on the same scale as other estimated parameters. This improves convergence of the HMC sampling of the posterior distribution.

[2] There are 71 total periods in the model period 1952-2022 and fishing mortality is not estimated in the terminal year.

3.1.3.1 Catch Treatment

Annual fishing mortality is estimated directly $F_t \sim N^+(0, \sigma_F)$ with catch fitted using a lognormal likelihood with uncertainty $\sigma_C = 0.2$.

Removals (catch) are calculated as:

$$\text{Removals}_{t-1} = \left(x_{t-1} + \text{surplus production}_{t-1}\right) \times \epsilon_t \times (1 - \exp(-F_{t-1})) \times \exp(\log K)$$

In the population dynamics model fishing mortality is applied through the exponential decay function $\exp(-F_{t-1})$ to determine the surviving population fraction, here removals are calculated as the complementary fraction $(1 - \exp(-F_{t-1}))$ multiplied by the total production (including surplus production and process error) and converted to absolute numbers units through $\exp(\log K)$.

The model fits to the observed total catch in numbers using a Student's t-distribution likelihood:

$$Removals_{obs,t} \sim \text{Student-t}\left(\nu_c, \log(Removals_t) - \frac{\sigma^2_{C,t}}{2}, \sigma_{C,t}\right)$$

where the degrees of freedom parameter ν_c controls the tail behavior, and the observation error $\sigma_{C,t}$ allows for time-varying uncertainty in catch observations. However, in the baseline model configuration $\sigma_{C,t} = 0.2$ for all t.

3.2 Prior Development

3.2.1 Biological Simulation Framework

Informative priors for key population parameters were developed through biological simulation using Monte Carlo sampling from biologically plausible parameter distributions. The simulation framework incorporated uncertainty and correlations in life history parameters:

- **Growth parameters**: Francis parameterization with independent sampling
- **Natural mortality**: Reference mortality M_{ref} negatively correlated with maximum age (r = -0.3)
- **Maturity**: Length-based logistic function parameters
- **Reproduction**: Steepness parameter for stock-recruit relationship and sex ratio
- **Weight-length relationship**: Correlated allometric parameters (r = -0.5) with biological constraints
- **Fishery selectivity**: Length at first capture for depletion prior

Parameter combinations (Table 1) were filtered to ensure biological realism and included correlations between life history parameters reflecting biological trade-offs (e.g., shorter lived individuals having higher natural mortality rates).

3.2.2 Maximum Intrinsic Rate of Increase (R_{Max}) Prior

R_{Max} was calculated by numerically solving the Euler-Lotka equation using bounded optimization (nlminb function in R) to find the value of R_{Max} that satisfies the equation within convergence criteria. Starting values and bounds were set to ensure biologically realistic solutions. Formulation of the Euler-Lotka equation followed the approach implemented in openMSE (Hordyk et al., 2024; Stanley et al., 2009):

$\alpha \sum_{a=1}^{A_{Max}} l_a f_a \exp(-R_{Max} \times a) = 1$

where α is the slope at the origin of the stock-recruitment curve, l_a is the probability of survival to age a, and f_a is the fecundity (reproductive output) at age a. Additional technical detail can be found in Section 11.1 of the Technical Annex.

Repeating this calculation across the 500,000 parameter combinations resulted in a prior distribution of R_{Max} conditioned on plausible biology and life-history characteristics. The resulting distribution was first filtered to values of $R_{Max} < 1.5$ representative of what the literature (Gravel et al., 2024; Hutchings et al., 2012) indicates is most likely for teleosts. A prior pushforward analysis, similar to (ISC, 2024), was used to further refine this distribution and develop a lognormal prior for R_{Max}. Briefly, random values of R_{Max} along with random values of the other leading parameters of the BSPM (drawn from the prior distributions described in Table 2) were used to drive simulated populations forward subject to direct removal of the observed catch (Figure 1). Based on the results of the prior pushforward (Figure 3), the R_{Max} distribution was further filtered based on the following criteria:

- Depletion in the terminal year > 2% and $R_{Max} < 1$
- Depletion in the terminal year > 2%, $R_{Max} < 1$, and trend in depletion over the last decade between -5% and 10%

This last filtering step approximates the recent relative trend seen in the index. Fitting a lognormal distribution to the remaining R_{Max} values results in the prior distribution specified in Table 2 and shown in Figure 4. Note that though initial filtering restricted $R_{Max} < 1$, the final lognormal prior still assigns some prior weight to $R_{Max} > 1$.

3.3 Model Implementation

The models were implemented using the No-U-Turn Sampler (NUTS) in Stan. Each model was run using 5 independent Markov chains with randomly generated starting values to ensure robust exploration of the posterior parameter space. Each chain was sampled for 3,000 iterations, with the first 1,000 iterations discarded as warmup to allow for algorithm adaptation. To reduce posterior sample autocorrelation and storage requirements, every 10^{th} samples was retained from the post-warmup iterations, yielding a final sample size of 1,000 draws from the joint posterior distribution across all chains. The NUTS algorithm was configured with strict adaptation parameters to ensure reliable convergence. The adaptation delta was set to 0.99 to reduce the step size and minimize divergent transitions, while the maximum tree depth was increased to 12 to allow for longer trajectories during the dynamic sampling process.

Model convergence and performance were assessed using multiple diagnostic criteria recommended for Bayesian stock assessment workflows (Monnahan, 2024). The potential scale reduction factor ($\hat{R}$) was required to be less than 1.01 for all parameters, indicating convergence across chains. Effective sample sizes were required to exceed 500 to ensure adequate precision in posterior estimates. Additionally, posterior predictive checks were conducted to evaluate model adequacy through comparison of observed versus predicted index and catch values, while parameter identifiability was assessed through visual examination of posterior distributions and correlation structures among estimated parameters (Gabry et al., 2019).

3.4 Model Diagnostics

Model performance was comprehensively evaluated using multiple diagnostic criteria recommended for Bayesian stock assessment workflows (Monnahan, 2024). The diagnostic framework included Stan model convergence criteria, data fits, posterior predictive checks, retrospective analysis, and hindcast cross-validation.

3.4.1 Convergence Diagnostics

Conventional Stan model convergence diagnostics and thresholds were employed to identify whether posterior distributions were representative and unbiased (Monnahan, 2024). Models were considered to have 'converged' to a stable, unbiased posterior distribution if they satisfied the following criteria:

- **Potential scale reduction factor** ($\hat{R}$) less than 1.01 for all leading model parameters, indicating convergence across chains
- **Bulk effective sample size** (N_{eff}) greater than 500 for all leading model parameters, ensuring adequate precision in posterior estimates

- **No divergent transitions** during the NUTS sampling process, which can indicate problematic posterior geometry
- **Maximum tree depth** not exceeded during sampling, ensuring adequate exploration of parameter space

Additionally, trace plots and rank plots were examined to visually assess mixing and convergence behavior across chains (Gabry et al., 2019).

3.4.2 Data Fits

Model fit to the abundance index was quantified using normalized root-mean-squared error (NRMSE):

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n}}$$

$$NRMSE = \frac{RMSE}{\sum_{i=1}^{n} y_i/n}$$

where y_i are the observed values and $\hat{y}_i$ are the model predictions. The average NRMSE across all posterior samples was calculated for each data component. Additionally, Probability Integral Transform (PIT) style residuals were calculated for the fit to the index.

3.4.3 Posterior Predictive Checks

Posterior predictive checks were conducted to evaluate model adequacy by comparing observed data with data simulated from the fitted model. For each posterior sample, synthetic datasets were generated using the estimated parameters, and agreement between observed and predicted datasets was assessed visually.

3.4.4 Retrospective Analysis

Retrospective analysis was conducted for each model by sequentially peeling off a year from the terminal end of the fitted index and re-running the model. Data were removed for up to five years, from 2022 back to 2018. Estimates of x in the terminal year of each retrospective peel were compared to the corresponding estimate of x from the full model run to better understand any potential biases or uncertainty in terminal year estimates. The Mohn's ρ statistic (Mohn, 1999) was calculated and presented. This statistic measures the average relative difference between an estimated quantity from an assessment (e.g., depletion in final year) with a reduced time-series of information and the same quantity estimated from an assessment using the full time-series.

Additionally, following Kokkalis et al. (2024) the proportion of retrospective peels (or coverage) where the relative exploitation rate (U/U_{MSY}) and relative depletion (D/D_{MSY}) were inside the credible intervals of the full model run was calculated. This metric provides an additional perspective on retrospective consistency by evaluating whether the uncertainty estimates from the full model appropriately capture the range of estimates obtained when using reduced datasets.

3.4.5 Hindcast Cross-Validation

Hindcast cross-validation (Kell et al., 2021) was conducted for each index to determine model performance in predicting the observed CPUE I one-step-ahead into the future relative to a naïve predictor. Briefly, the 'model-free' approach to hindcast cross-validation was used, and used the same set of five retrospective peels described in Section 3.4.4.

The 'model-free' hindcast calculation is described using the model from the last peel $BSPM_{2017}$ as an example. This model fit to index data through 2017 but included catch through 2022. The model estimates of predicted CPUE in 2018 based on $BSPM_{2017}$ is the 'model-free' hindcast for 2018, $\hat{I}_{2018}$. The naïve prediction of CPUE in 2018 is simply the observed CPUE from 2017, $I_{2017} = \ddot{I}_{2018}$. The absolute scaled error (ASE) of the prediction is:

$ASE_{2018} = \frac{|I_{2018} - \hat{I}_{2018}|}{|I_{2018} - \ddot{I}_{2018}|}$

Repeating this calculation across all retrospective peels for years 2019-2022 and taking the average across ASE values gives the mean ASE or MASE for the model. An MASE value less than one indicates that the model has greater predictive skill than the naïve predictor.

3.5 Forecasts

Simple, status quo catch-based stochastic projections over a 10-year window are provided for each model. For each model, projected catch was taken as the average catch over the terminal 5 years (2018-2022). Process error in the projection period was randomly resampled from the estimated process error in the main model period.

4 Results

4.1 Model Convergence and Diagnostics

All three BSPM model variants achieved satisfactory convergence based on standard Hamiltonian Monte Carlo diagnostics (Table 3). All models met convergence criteria with $\hat{R}$ values well below the 1.01 threshold and effective sample sizes exceeding the recommended minimum of 500 (100 samples per chain). No divergent transitions were observed across chains, and maximum tree depth was not exceeded during sampling, indicating successful exploration of the posterior space.

4.1.1 Model Fit to Data

4.1.1.1 Catch Data Fit

All models fitted the observed catch time series (1952-2022) with a coefficient of variation of 0.2 to account for uncertainty in catch estimates. Since fishing mortality required to generate the catch was directly estimated, model fit to the catch was good (Figure 10). The models consistently predicted lower than observed catch for the large catch event in 1954.

4.1.1.2 CPUE Index Fit

Model fit to the standardized CPUE index (1988-2022) was evaluated using normalized root-mean-squared error (NRMSE). All three models provided reasonable fits to the observed index given the high variability and observation error (Table 3). The `0003-2024cpueFPrior` model showed marginally improved fit with lower RMSE values. All models successfully captured the general declining trend in standardized CPUE from 1988-2022. However, the estimated combination of observation and process error meant models were not tightly constrained to individual observations, resulting in persistent negative residuals in the last decade (Figure 11). Posterior predictive checks confirmed that models replicated key features of the observed CPUE data (Figure 12).

4.1.2 Model Validation

Overall retrospective bias was very low and close to zero for all models (Table 3; Figure 13), indicating lack of persistent bias in terminal estimates as successive years of index data were removed. Estimates of relative exploitation rate (U/U_{MSY}) and relative depletion (D/D_{MSY}) fell within the credible intervals of the full model run 100% of the time (Table 3). For hindcast cross-validation, all models exceeded the performance of the naïve predictor, with the `0003-2024cpueFPrior` model showing the best performance (Table 3; Figure 14). This provides support that models successfully identified and estimated a production function for the stock.

4.2 Population Dynamics and Stock Status

In all models, most leading parameters (*log(K)*, R_{Max}, σ_P, n, σ_F, and $\sigma_{O,add}$) showed posterior updates from their respective prior distributions (Figure 15), indicating sufficient information in the data to inform parameter estimates. The shape parameter n showed little deviation from the Schaefer-model prior. All models estimated similar levels of σ_P and additional observation error $\sigma_{O,add}$ (Table 4), which is unsurprising given they fitted the same catch and standardized index. Although estimates of σ_P were elevated from the prior, models primarily reconciled index variability by estimating substantial additional observation error $\sigma_{O,add}$, resulting in approximately a 3:1 ratio of total observation error σ_O to process error σ_P. Similar to findings in ISC (2024), there was a trade-off between *log(K)* and σ_F where higher population scale was associated with lower fishing mortality and vice versa.

All models showed substantial posterior updates for the R_{Max} parameter, estimating productivity on the lower end of the biological prior. These median estimates of R_{Max}, between 0.2-0.3 (Table 4), are consistent with estimates for an analogous species in the Atlantic Ocean, white marlin (*Kajikia albida*). The most recent ICCAT assessment (ICCAT, 2020), implemented using JABBA (Winker et al., 2018), estimated R_{Max} at 0.163 (95% credible interval: 0.122-0.215), which falls within the posterior estimates of all three models (Table 4). Notably, the Atlantic white marlin JABBA model showed remarkable consistency with stock status and trend estimates from a fully integrated, length-structured model implemented in Stock Synthesis (ICCAT, 2020). In the current analysis, the `0003-2024cpueFPrior` model estimated lower productivity R_{Max} with reduced uncertainty (Table 4), consistent with assuming a more depleted stock. Inverse transform sampling to match the R_{Max} posterior distribution from this model reveals that lower productivity is characterized by pronounced shifts toward lower steepness values h and, to a lesser extent, smaller von Bertalanffy k values (Figure 16).

The estimated population depletion time series shows a decline from the assumed unfished state in 1952 to minimum levels around the early 1990s (Figure 17). All three models estimated similar depletion trajectories, with differences primarily related to estimated population scale. The `0003-2024cpueFPrior` model estimated slightly lower depletion levels throughout the time series compared to the other models. From the minimum depletion period, all models estimated gradual recovery through the 2000s and 2010s, with populations stabilizing at moderate depletion levels in recent years. Credible intervals around depletion estimates are relatively wide, particularly in the early period when no abundance index data were available to inform the model, reflecting inherent uncertainty in reconstructing historical population dynamics. The estimated population time series mirrors these depletion patterns. The `0001-2024cpueExPrior` model estimated the highest population scale throughout the time series, while the `0002-2024cpueDepPrior` and `0003-2024cpueFPrior` models estimated lower population levels. While uncertainty in absolute population size is substantial, as reflected in wide credible intervals, relative trends are more precisely estimated.

Posterior estimates of depletion in 1988 differed notably among the three models (Table 4). For the `0002-2024cpueDepPrior` and `0003-2024cpueFPrior` models, these estimates differed from the prior value of $x_{1988} = 1$ (Table 2). The baseline model (`0001-2024cpueExPrior`) estimated median depletion of $x_{1988} = 0.87$ (0.59-1.06), while the depletion-constrained model (`0002-2024cpueDepPrior`) estimated $x_{1988} = 0.75$ (0.56-0.94). The `0003-2024cpueFPrior` model, with a more representative prior on fishing mortality variability, estimated the greatest depletion levels in 1988: $x_{1988} = 0.67$ (0.50-0.89).

Process error estimates (Figure 17) showed consistent temporal variability across models, with the largest deviations occurring during periods of rapid population change in the 1970s and 1980s. This indicates that removals alone could not explain the observed changes in the standardized index. The estimated process error variability σ_P (Table 4), while similar across all models, was lowest for the `0003-2024cpueFPrior` model, indicating greater internal consistency in the model dynamics.

4.3 Biological Reference Points

MSY-based reference points were derived from the Fletcher-Schaefer production function for all three BSPM variants. The relative stock status time series (Figure 17) shows consistent trends across all models, albeit at different relative levels. Median estimates of the D/D_{MSY} ratio indicate the stock was above the MSY reference level for the entire model period. Only the `0003-2024cpueFPrior` model indicates risk of falling below the MSY-based reference point in recent years. Similarly, median estimates of the F/F_{MSY} ratio were consistently below the overfishing reference point, with episodic periods of overfishing risk indicated by the `0003-2024cpueFPrior` model. However, recent trends indicate declining overfishing risk, consistent with estimated stock recovery.

4.4 Future Projections

Ten-year stochastic projections (2023-2032) were conducted for all three BSPM variants under a status quo catch scenario, with projected removals set to the average catch over 2018-2022 (Figure 18). Under status quo catch levels, all models projected continued population recovery through 2032. Depletion trajectories show gradual increases from current levels, with the `0001-2024cpueExPrior` model projecting the most optimistic recovery due to higher estimated population scale. The `0002-2024cpueDepPrior` and `0003-2024cpueFPrior` models showed more modest but consistent increases in depletion over the projection period.

Relative to MSY-based reference points, projections indicate the stock will likely remain above D_{MSY} throughout the projection period for the baseline and depletion-constrained models. The `0003-2024cpueFPrior` model showed greater variability and some risk of declining below D_{MSY} in the near term, though the median trajectory suggests stabilization above the reference point by the end of the projection period. Projected fishing mortality rates remained well below F_{MSY} for all models under the status quo catch scenario, indicating that current harvest levels are likely sustainable and consistent with continued stock recovery. Projection uncertainty increased over time, reflecting both parameter uncertainty and stochastic variability in future population dynamics.

5 Discussion

5.1 General remarks

The three BSPM variants provide a consistent picture of stock trajectory and current status. All models estimated similar depletion patterns: decline from unfished conditions in 1952 to minimum levels around the early 1990s, followed by gradual recovery through recent decades. This trajectory suggests the stock is currently recovering. While median estimates indicate the stock was not overfished or undergoing overfishing during the assessment period, the `0003-2024cpueFPrior` model suggests some recent risk of falling below these thresholds.

The BSPM approach addresses key limitations encountered in previous integrated assessments while providing robust stock status estimates. The primary advantage lies in condensing complex biological and fishery uncertainties into a manageable parameter set. Integrated assessments can be influenced by size composition data, which are sensitive to assumptions about growth, mortality, selectivity, and reproduction parameters. When these assumptions are uncertain or mis-specified, population scale estimates informed by size composition data can be biased. Production models circumvent this issue by aggregating these uncertainties into productivity parameters, allowing more efficient exploration of plausible population dynamics while maintaining biological realism through informed priors.

5.2 Challenges, limitations & key uncertainties

Several important limitations affect this analysis. The assumption of no age-specific population dynamics may be problematic when mortality or reproductive potential varies significantly by age. This simplification prevents incorporation of biological information beyond production function parameters and excludes explicit use of size composition data.

The aggregated nature of surplus production models can potentially bias MSY estimates when age-specific processes differ from production function assumptions. For species with complex life histories like striped marlin, managers should consider these limitations when applying MSY-based reference points derived from surplus production models.

Data limitations also constrain the analysis. Fitting to a single standardized abundance index limits evaluation of uncertainty in abundance trends. Multiple indices would enable model ensemble approaches to better characterize this uncertainty.

The incorporation of depletion information from New Zealand recreational size data through a 1988 depletion prior, while novel, represents a simplified approach. More sophisticated modeling frameworks like delay-difference or age-structured models could explicitly fit size composition time series in a likelihood-based framework, providing more elegant integration of this information.

Modeling assumptions introduce additional uncertainties. Current modeling assumes the same level of catch uncertainty across all time periods. Future versions could allow time-varying catch uncertainty (e.g., greater uncertainty in historical versus recent observations) or accommodate effort-driven removals similar to catch-errors with effort deviation approaches available in MULTIFAN-CL (Fournier et al., 1998).

Population scale estimates were highly uncertain and sensitive to model configuration choices, while depletion estimates were more precisely estimated. The choice of prior on σ_F was particularly influential. Although this prior was developed using prior pushforward analysis (Kim & Neubauer, 2025), it represents a key uncertainty. Similarly, uncertainty in biological parameters (growth, maturity, natural mortality) due to limited sample sizes contributed to broad initial priors for R_{Max}. While the posterior distribution for R_{Max} was substantially updated from the prior, reducing uncertainty in key biological productivity parameters could help reduce model uncertainty estimates and facilitate development of age-structured models.

5.3 Future stock assessment modeling considerations

Future assessment efforts should progressively build toward full age-structure through a systematic approach: delay-difference models, age-structured production models, and then fully integrated age-structured models. This development should occur within a Bayesian framework to inform parameter estimates where prior information exists or can be derived using prior pushforward approaches (Kim & Neubauer, 2025).

Developing Bayesian age-structured modeling approaches can provide more stable estimation of increased parameters due to the use of priors, explicitly account for parameter uncertainty rather than making fixed assumptions, potentially enable informed estimation of difficult-to-observe biological relationships (e.g., stock-recruit relationships), address BSPM limitations by explicitly considering age-structured processes and fishery selectivity, and estimate leading parameters with likelihood components for indices and size composition.

While simplified models offer advantages when data conflicts exist, over-simplification of age-structured processes could result in biases. Future modeling must balance complexity with data quality and conflicts. The transition should be guided by careful evaluation of whether increased model complexity is supported by available data and whether it improves the reliability of management advice.

6 Acknowledgements

The authors appreciate feedback received on earlier versions of this analysis, and are thankful to the following for their insights and thoughtful discussion which greatly improved the assessment: K. Kim (SPC), N. Davies (Te Takina Ltd.), S. Hoyle (Hoyle Consulting), R. Ahrens (PIFSC), M. Nadon (PIFSC), and F. Carvalho (PIFSC).

7 Declaration of Generative AI use

A generative artificial intelligence (AI) assistant, Anthropic Claude Sonnet 4.0, was used to parse model code in the preparation of an initial draft of this report.

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9 Tables

Table 1: Biological parameter distributions and sources used in simulation framework

Parameter	Symbol	Distribution	Mean/Mode	CV/Range	Correlation	Source/Notes
Length at age 1	L_1	Lognormal	60 cm	0.2	Independent	(Ducharme-Barth et al., 2025)
Length at age 10	L_2	Lognormal	210 cm	0.2	Independent	(Ducharme-Barth et al., 2025)
Growth coefficient	k	Beta(6.5, 3.5)	~0.65	-	Independent	(Ducharme-Barth et al., 2025)
Length CV	cv_{len}	Uniform	-	[0.05, 0.25]	-	Growth variability
Maximum age	A_{Max}	Lognormal	15 years	0.2	ρ = -0.3 with M_{ref}	(Farley et al., 2021)
Reference mortality	M_{ref}	Lognormal	0.36 yr^{-1}	0.44	ρ = -0.3 with A_{Max}	(Hamel & Cope, 2022) ($5.4/A_{Max}$)
Maturity slope	a_{mat}	Normal	-20	0.2	-	Logistic steepness
Length at 50% maturity	L_{50}	Lognormal	184 cm	0.2	-	(Farley et al., 2021)
Weight coefficient	a_w	Lognormal	5.4×10^{-7}	0.05	ρ = -0.5 with b_w	(Castillo-Jordan et al., 2024)
Weight exponent	b_w	Normal	3.58	0.05	ρ = -0.5 with a_w	(Castillo-Jordan et al., 2024) Constrained: 150-350 kg at 300 cm
Steepness	h	Beta(3, 1.5)	~0.73	-	-	Censored to [0.2, 1.0]
Sex ratio (prop. female)	s	Normal	0.5	0.05	-	Constrained [0.01, 0.99]
Reproductive cycle	-	Fixed	1 year	-	-	Annual spawning
Reference ages	age_1, age_2	Fixed	0, 10 years	-	-	Francis parameterization

Table 2: Prior distributions for effort-based Bayesian surplus production model parameters.

Parameter	Distribution	Mean	SD	Description	Key Correlations
Core Population Parameters					
K	Lognormal	$\log(1,110,533)$	0.59	Carrying capacity (N)	$\rho(K, R_{Max})$ = -0.22; $\rho(K, q_{eff})$ = -0.241
R_{Max}	Lognormal	log(0.6206)	0.58	Maximum intrinsic rate of increase	$\rho(R_{Max}, n)$ = -0.881
n	Lognormal	log(1.16)	0.54	Pella-Tomlinson shape parameter	
x_0	Lognormal	log(1)	0.025	Initial depletion	(Independent)
q_{eff}	Lognormal	log(0.5019)	0.95	Effort catchability coefficient	$\rho(q_{eff}, \sigma_{qdev}) = 0.559$; $\rho(q_{eff}, \rho) =$ 0.243
ρ	Transformed Normal	$\tanh^{-1}(0.981)$	0.62	Autocorrelation in catchability deviations	$\rho(\rho, \sigma_{qdev}) = 0.148$
σ_{qdev}	Lognormal	log(1.619)	0.3	Catchability variability standard deviation	
Process and Observation Error					
σ_P	Lognormal	log(0.0533)	0.27	Process error standard deviation	(Independent)
$\sigma_{O,add}$	Half-Normal	0	0.2	Additional estimated observation error	(Independent)
Catch Likelihood					
ν_{catch}	Gamma	Shape = 2	Rate = 0.1	Student-t degrees of freedom for catch	(Independent)

Table 3: Model summary

Model	Max $\hat{R}$	Min ESS	Divergent	Tree Depth	BFMI Issues	Index RMSE	Mohn's ρ	MASE	Coverage D/D_{MSY}	Coverage U/U_{MSY}
`0001`	1.009	724	0	0	0	0.28	0.006	0.966	100%	100%
`0002`	1.009	674	0	0	0	0.27	-0.005	0.863	100%	100%
`0003`	1.008	728	0	0	0	0.265	-0.026	0.779	100%	100%

Notes:

`0001` corresponds to the `0001-2024cpueExPrior` model

`0002` corresponds to the `0002-2024cpueDepPrior` model

`0003` corresponds to the `0003-2024cpueFPrior` model

- Max $\hat{R}$: Maximum potential scale reduction factor across all parameters (should be < 1.01)
- Min ESS: Minimum effective sample size across all parameters (should be > 500)
- Divergent: Number of divergent transitions during sampling (should be 0)
- Tree Depth: Whether maximum tree depth was exceeded during sampling
- BFMI Issues: Whether low Bayesian Fraction of Missing Information was detected
- Index RMSE: Root mean squared error for fit to standardized CPUE index
- Mohn's ρ: Retrospective bias statistic (values close to 0 indicate low bias)
- MASE: Mean Absolute Scaled Error from hindcast cross-validation (< 1 indicates model outperforms naïve predictor)
- Coverage D/D_{MSY}: Percentage of retrospective peels where relative depletion estimates fall within full model credible intervals
- Coverage U/U_{MSY}: Percentage of retrospective peels where relative exploitation rate estimates fall within full model credible intervals

Table 4: Posterior estimates with 95% credible intervals for all BSPM parameters

Model	*log(K)*	R_{Max}	σ_P	*Shape (n)*	σ_F	$\sigma_{O,add}$	σ_O	x_{1988}
`0001`	13.8 (13.2-14.6)	0.27 (0.13-0.76)	0.07 (0.04-0.11)	1.97 (1.62-2.37)	0.03 (0.01-0.07)	0.10 (0.04-0.19)	0.27 (0.20-0.35)	0.87 (0.59-1.06)
`0002`	13.7 (13.2-14.5)	0.23 (0.12-0.56)	0.07 (0.04-0.12)	1.96 (1.62-2.35)	0.04 (0.02-0.08)	0.09 (0.03-0.17)	0.26 (0.20-0.34)	0.75 (0.56-0.94)
`0003`	13.3 (12.8-14.0)	0.23 (0.13-0.45)	0.06 (0.03-0.10)	1.97 (1.65-2.38)	0.07 (0.03-0.13)	0.09 (0.03-0.16)	0.26 (0.19-0.33)	0.67 (0.50-0.89)

Parameter Definitions:

`0001` corresponds to the `0001-2024cpueExPrior` model

`0002` corresponds to the `0002-2024cpueDepPrior` model

`0003` corresponds to the `0003-2024cpueFPrior` model

- *log(K)*: Natural logarithm of carrying capacity (total numbers)
- R_{Max}: Maximum intrinsic rate of population increase (per year)

- σ_P: Process error standard deviation
- *Shape (n)*: Pella-Tomlinson production function shape parameter
- σ_F: Fishing mortality variability standard deviation
- $\sigma_{O,add}$: Extra estimated observation error component
- σ_O: Total observation error (input + additional)
- x_{1988}: Population depletion in 1988 (year 37 of model period)

10 Figures

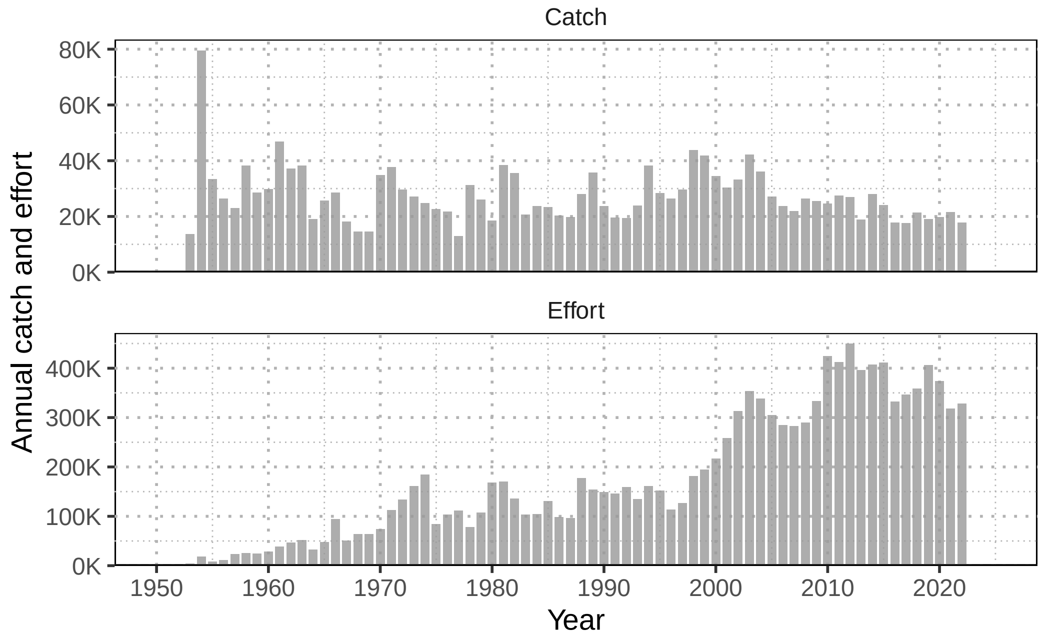

Figure 1: Annual catch (numbers; individuals) of striped marlin and nominal longline effort (hooks fished; thousands) in the Southwest Pacific Ocean (1952-2022).

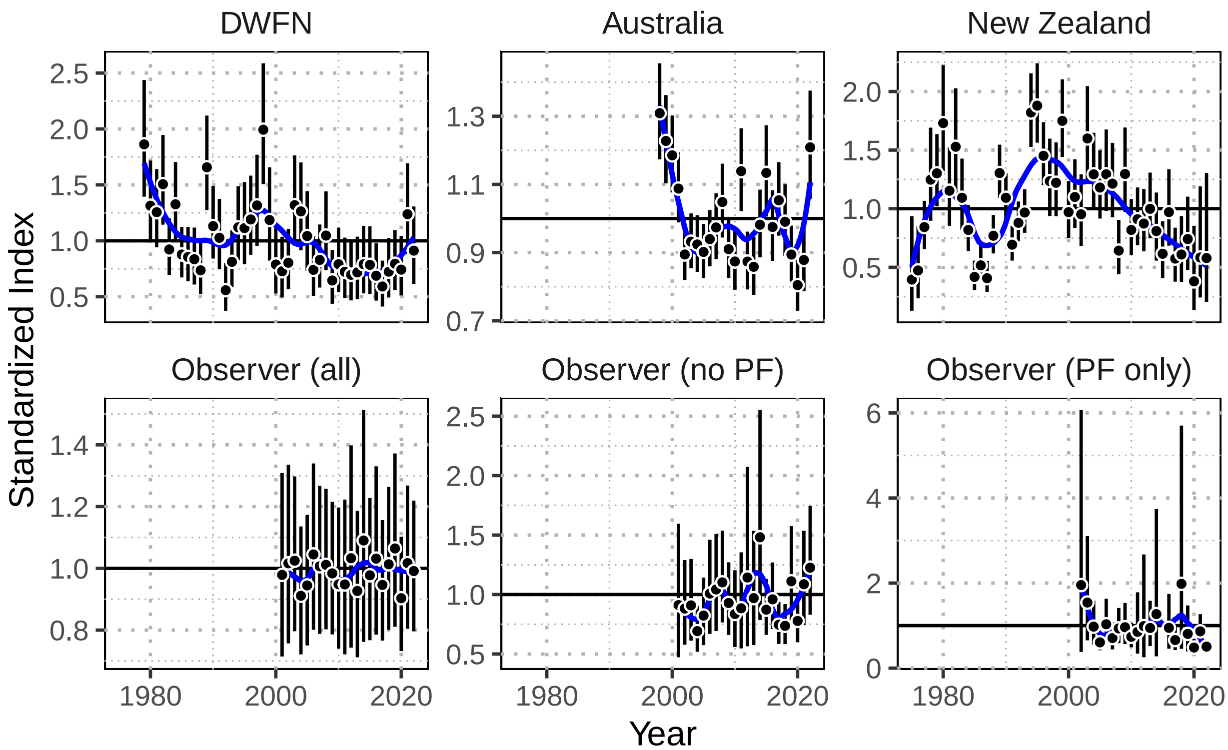

Figure 2: Standardized index (black points), and observation error (black bars) for alternative model indices. The trend in the indices is shown in blue using a loess-smooth.

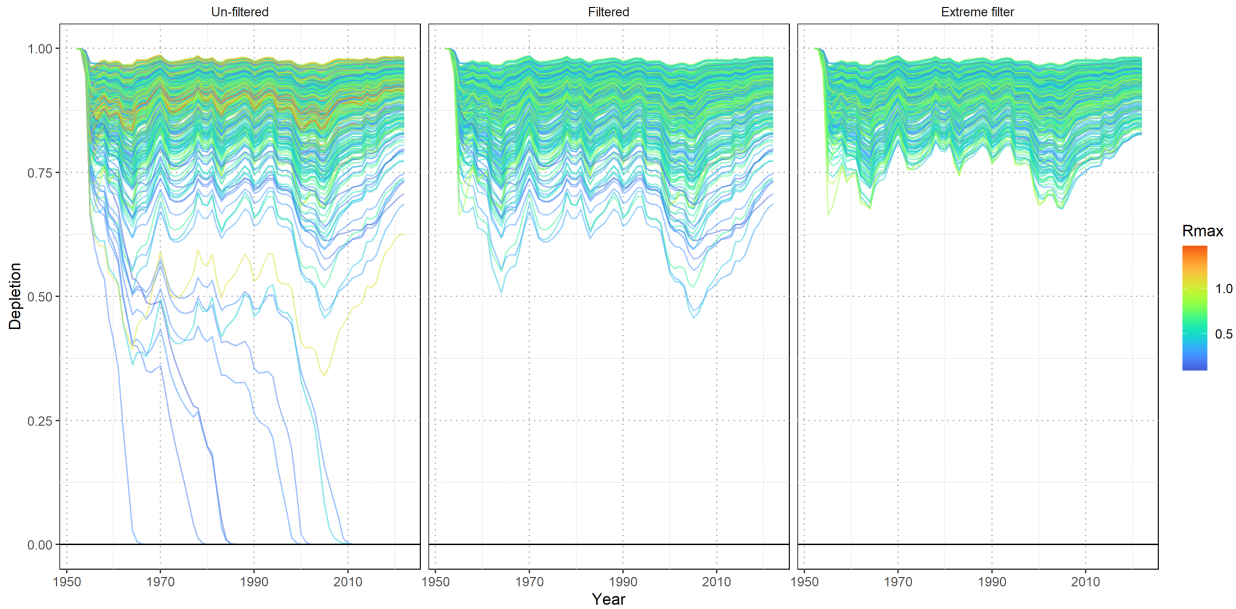

Figure 3: Prior pushforward check showing simulated population depletion trajectories under different filtering scenarios. Lines represent population trajectories colored by maximum intrinsic rate of increase (R_{Max}). Panels show progressively stricter filtering criteria from left to right.

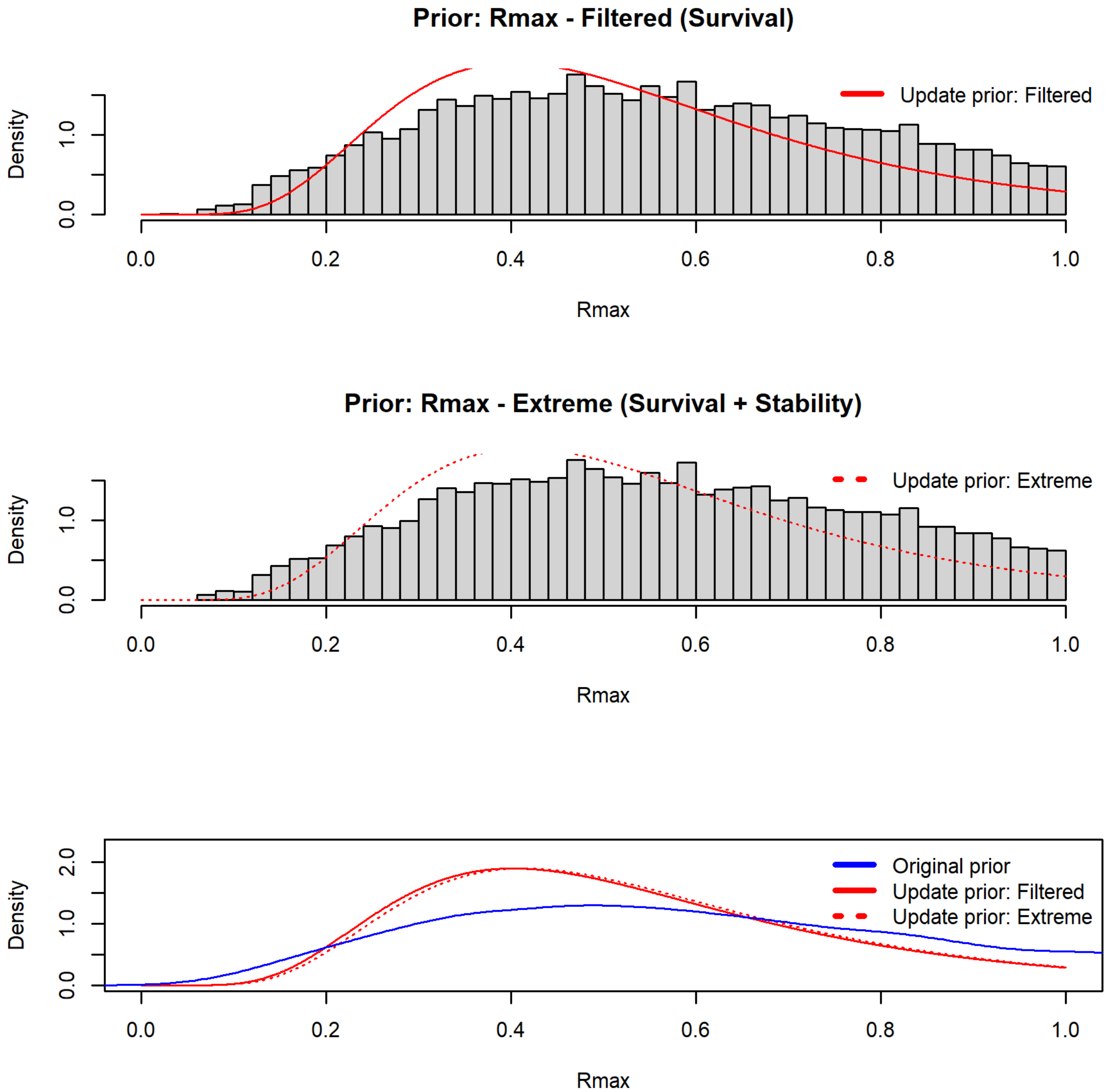

Figure 4: Prior distributions for maximum intrinsic rate of population increase (R_{Max}). Upper panel shows baseline filtered values (gray histogram) with fitted lognormal distribution (red line). Middle panel shows extreme filtered values with fitted distribution (dotted red line). Lower panel compares original simulation values (gray), viable populations (blue), and final lognormal priors (red).

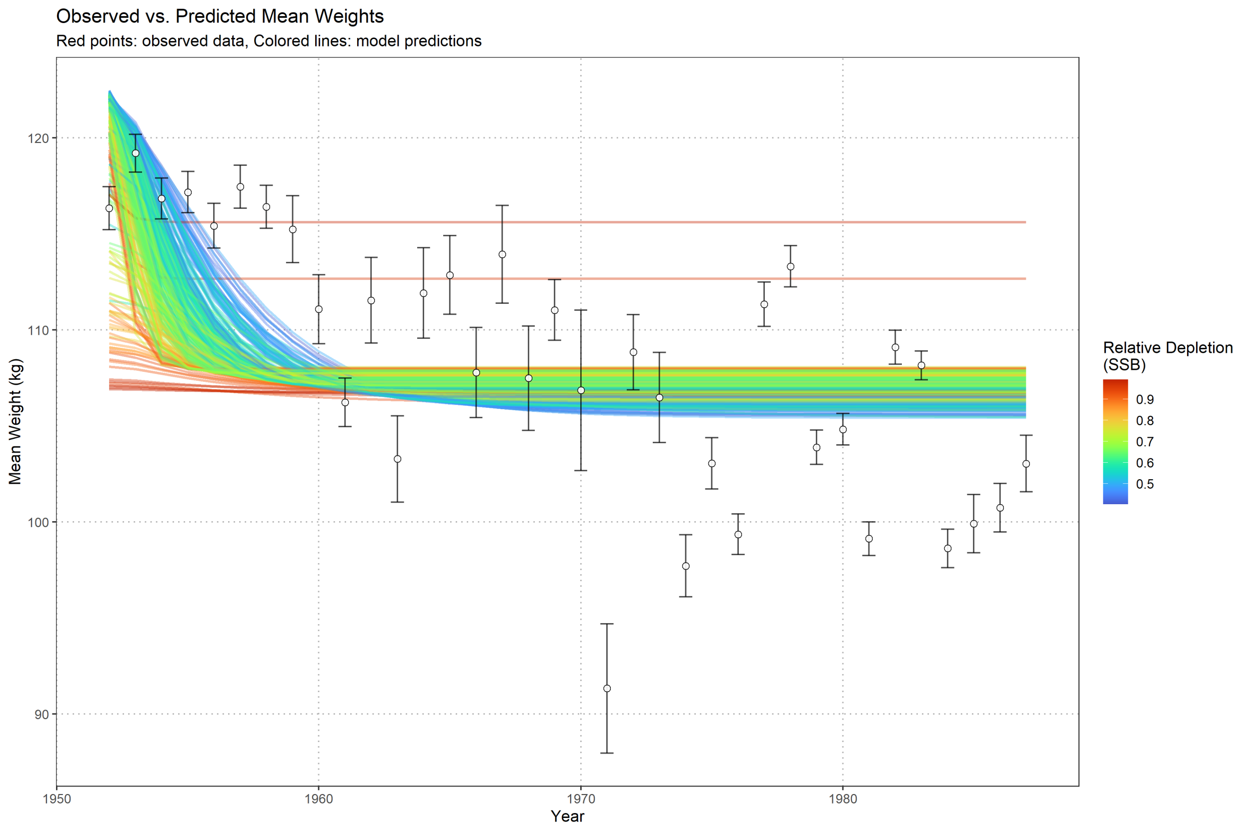

Figure 5: Observed versus predicted mean weights from New Zealand recreational fishery data. Points show observed mean weights with error bars, colored lines show model predictions from the transitional mean weight model colored by relative spawning stock biomass (SSB) depletion.

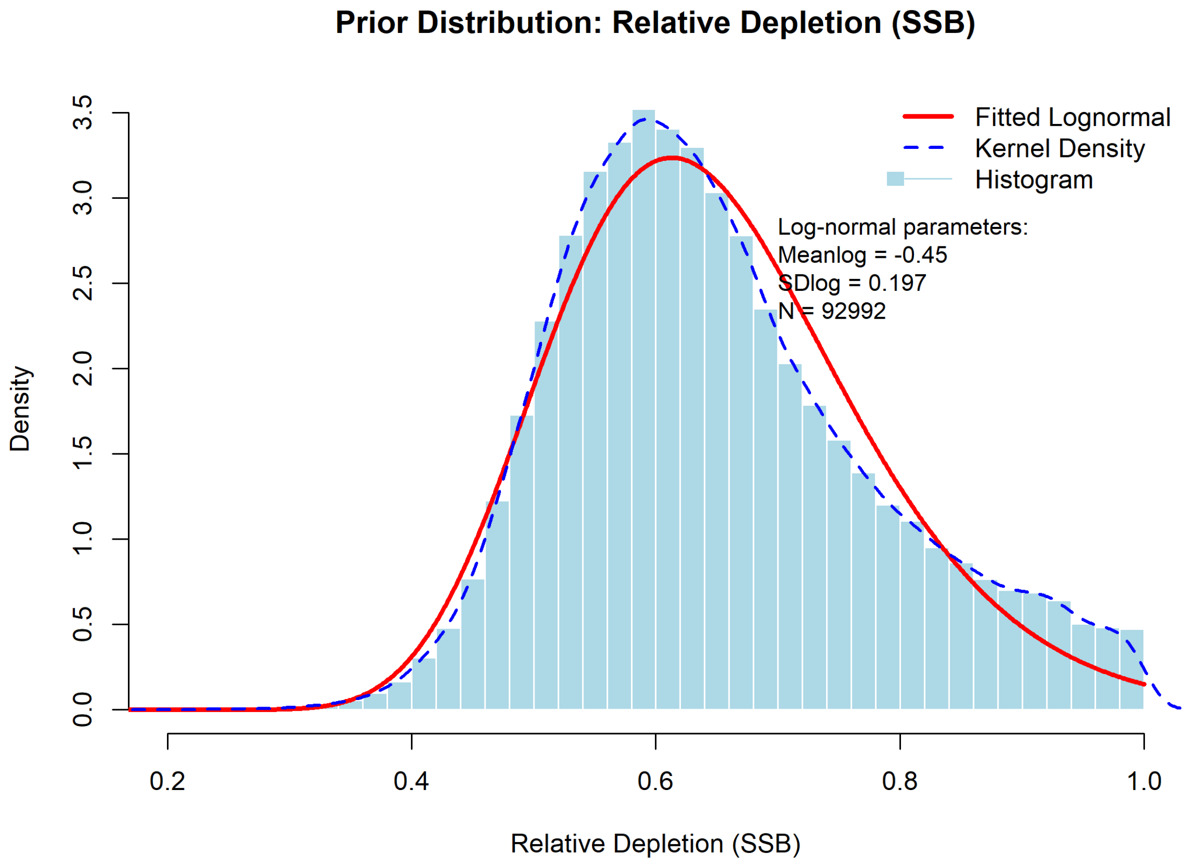

Figure 6: Prior distribution for relative depletion in 1988 derived from New Zealand recreational weight data. Light blue histogram shows depletion estimates from biological parameter combinations that successfully fit the transitional mean weight model. Blue dashed line shows kernel density estimate, red line shows fitted lognormal distribution used as prior in the depletion-constrained BSPM.

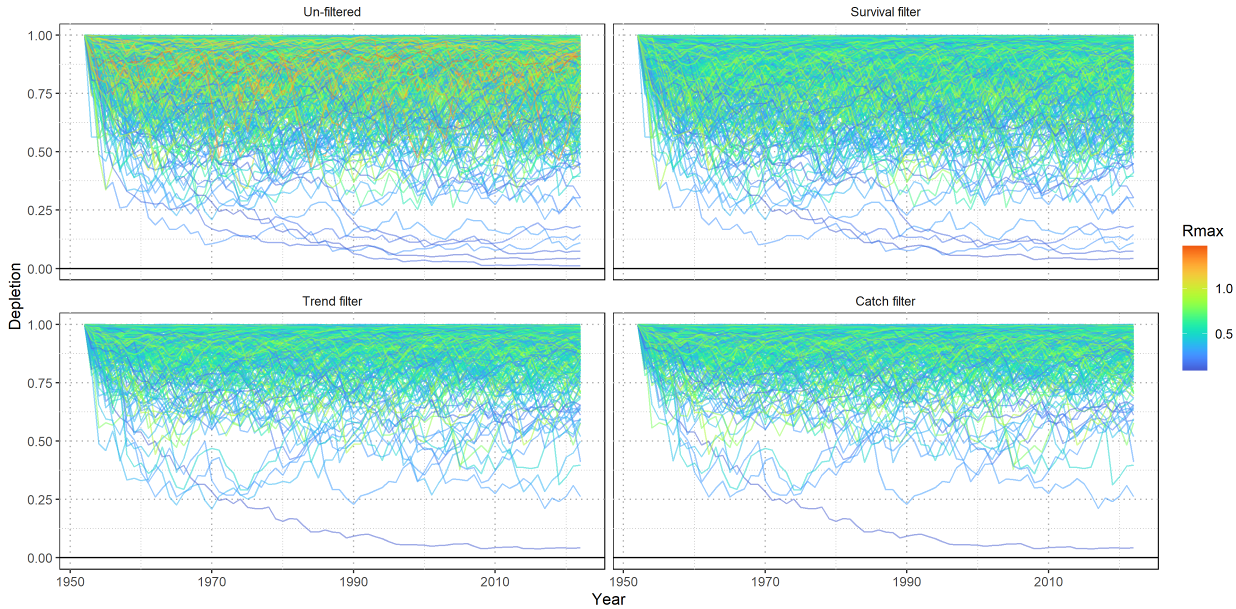

Figure 7: Prior pushforward check for population depletion trajectories under different filtering scenarios. Each line represents a simulated population trajectory colored by fishing mortality variability (σ_F).

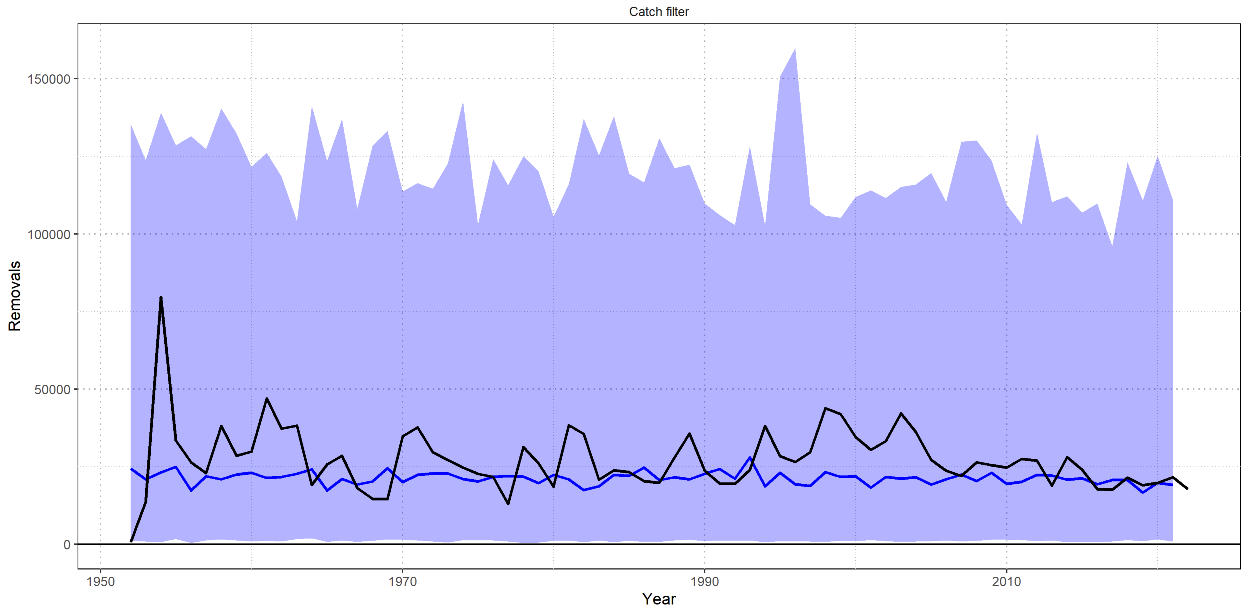

Figure 8: Simulated removals from the fishing mortality variability (σ_F) prior pushforward analysis. Solid blue line indicates the median simulated removals with 95^{th} percentile (shaded region). Observed removals are shown by the solid black line.

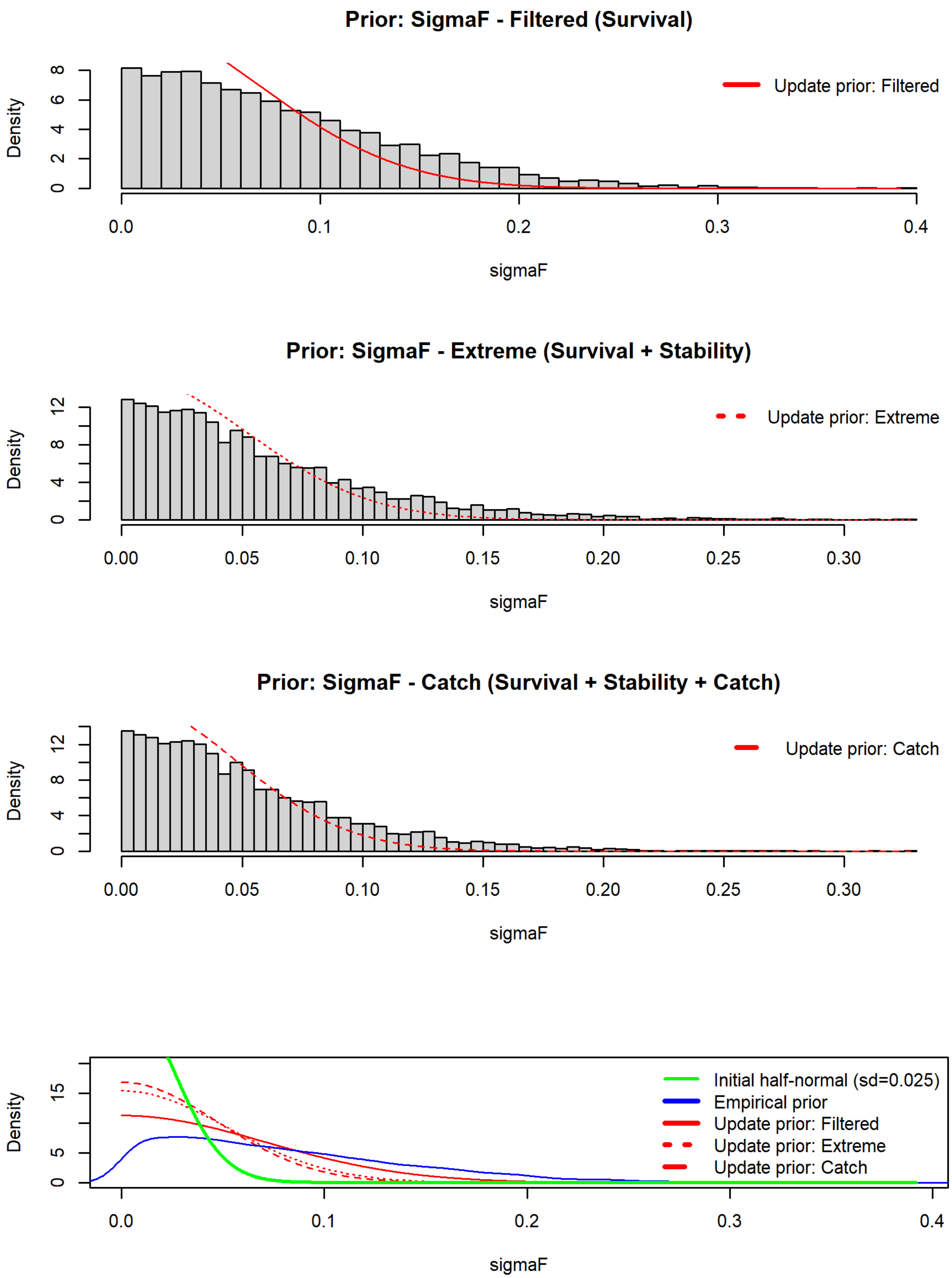

Figure 9: Prior distributions for fishing mortality variability (σ_F) under different filtering scenarios. Top three panels show progressively stricter filtering (baseline, extreme, catch-based) with fitted distributions (red lines). Bottom panel compares all distributions: original simulation (gray), viable populations (blue), final priors (red), and prior used in models `0001` and `0002` (green).

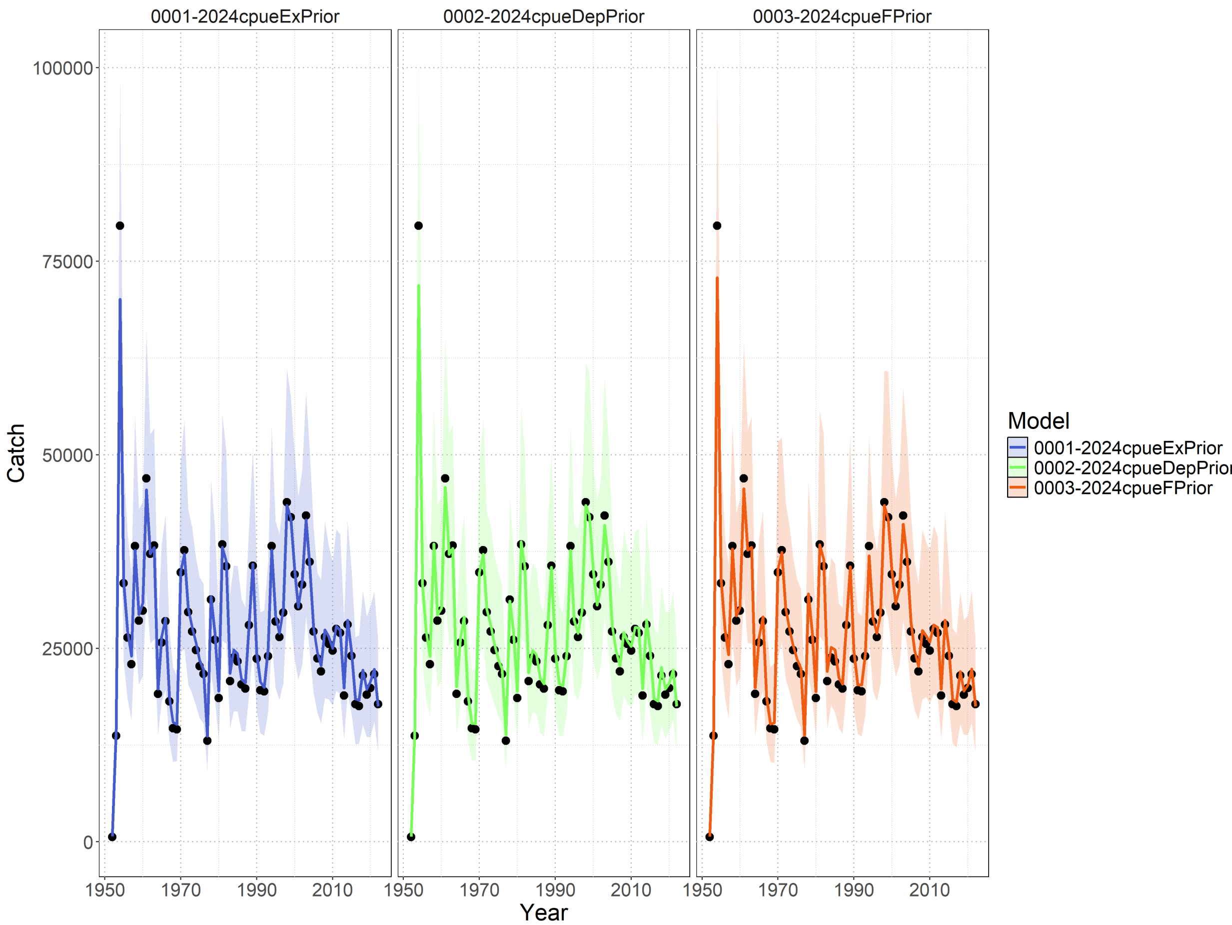

Figure 10: Time series of observed catch (black dots with uncertainty bounds) and model-predicted catch for three Bayesian surplus production models fitted to SWPO striped marlin data from 1950-2022. The left panel shows model `0001-2024cpueExPrior` (baseline model with expert priors), the central panel shows model `0002-2024cpueDepPrior` (depletion-constrained model incorporating 1988 depletion prior from New Zealand recreational weight data), the right panel shows model `0003-2024cpueFPrior` which assumes a less restrictive prior on fishing mortality variability. Colored lines represent posterior median predictions with shaded uncertainty bands showing 95% credible intervals.

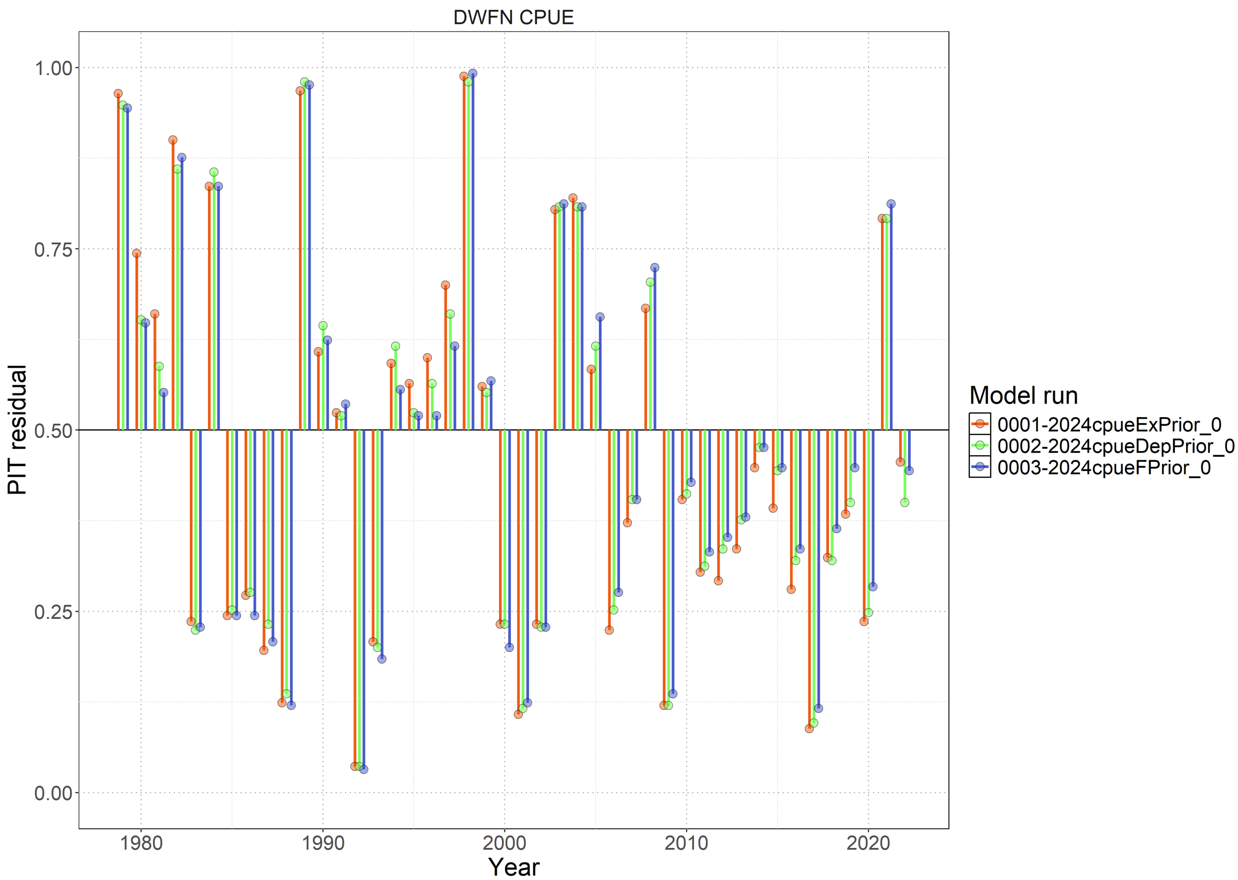

Figure 11: Probability-integral transform (PIT) style residuals from each model (colored points) fit to the standardized index.

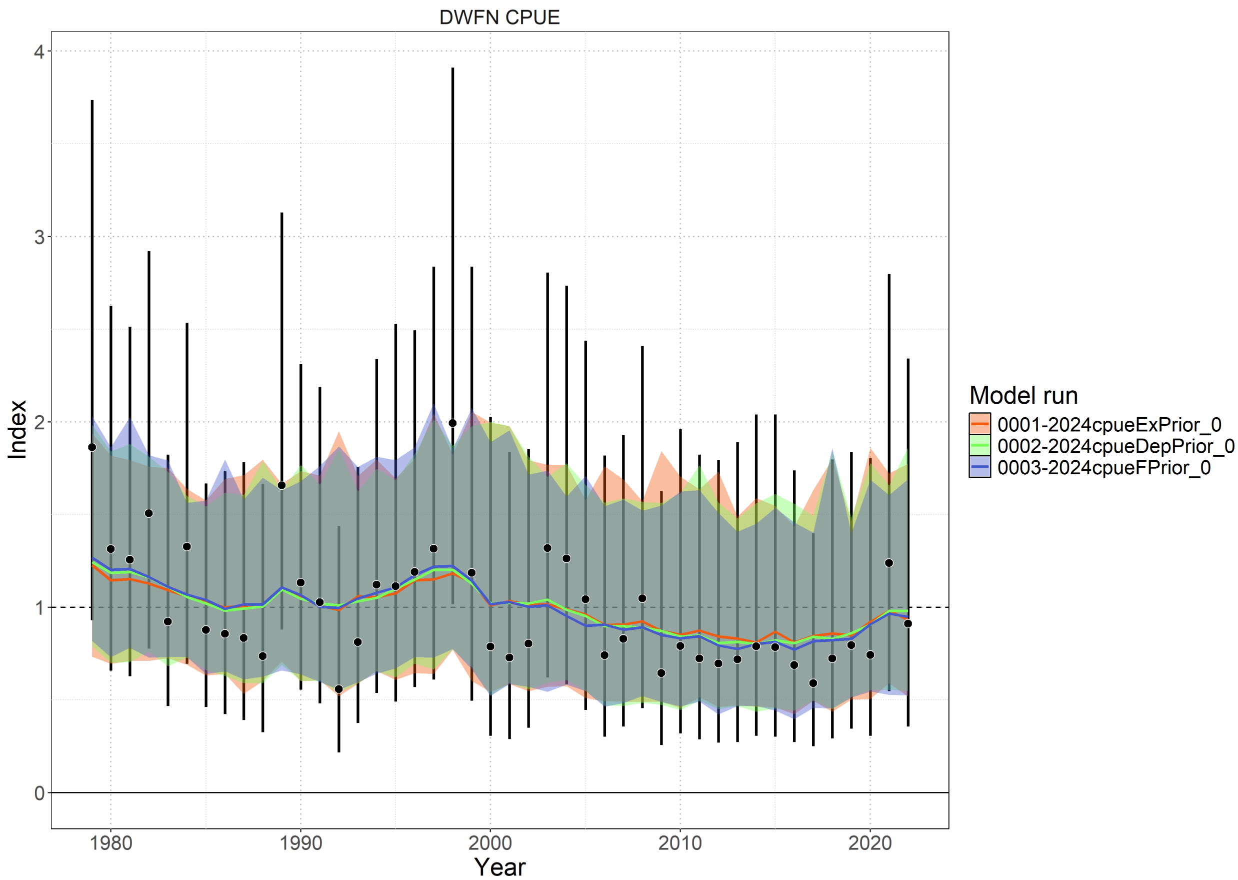

Figure 12: Standardized index (black points), observation error (black bars), and posterior predicted model fits (colored lines) with associated 95% credible interval (colored shading).

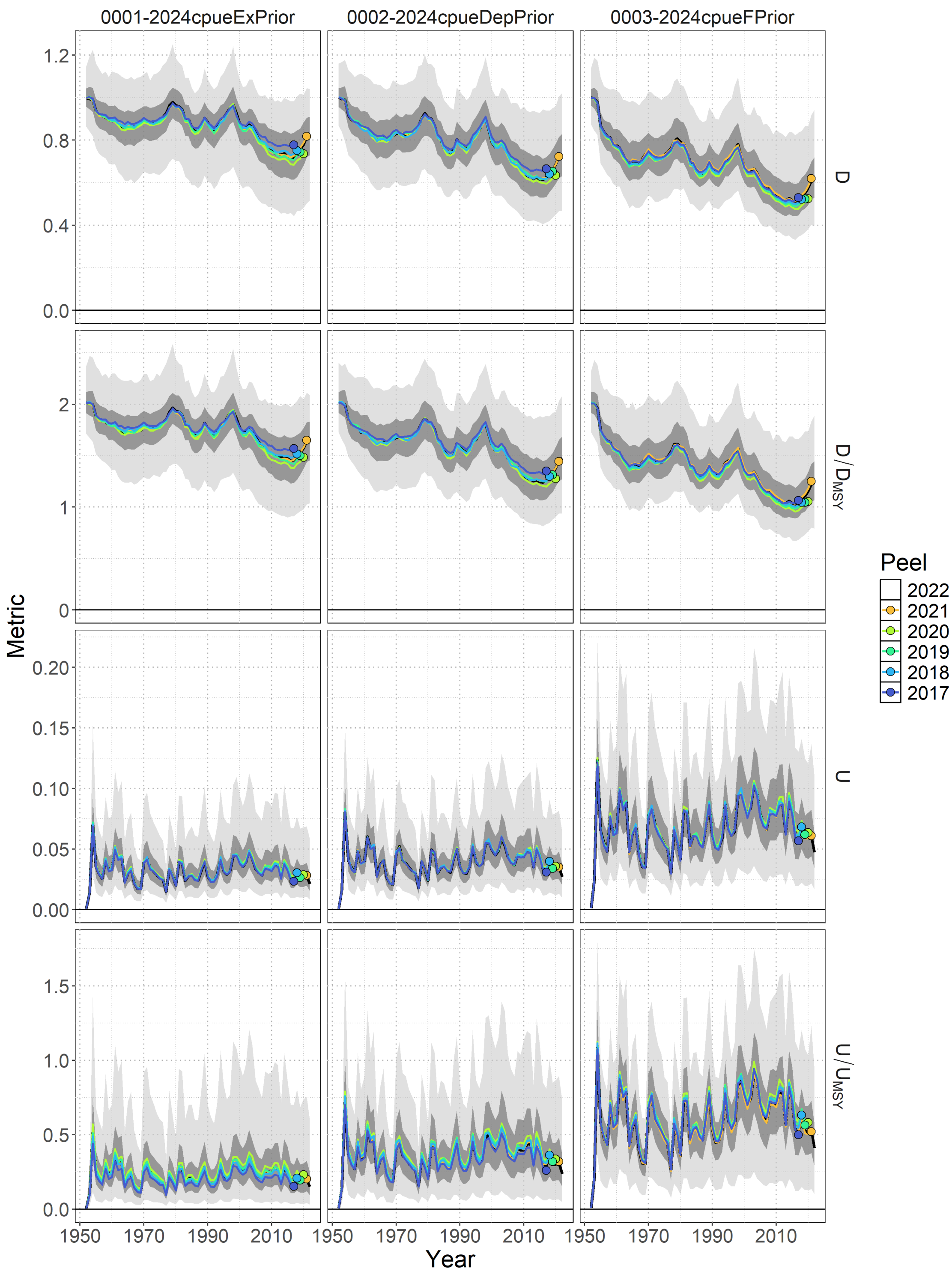

Figure 13: Retrospective analysis for all models with respect to time series of depletion D_t, depletion relative to depletion at MSY D_t/D_{MSY}, exploitation rate U_t, and exploitation rate relative to the rate of exploitation that produces MSY U_t/U_{MSY}. The base model with data included through 2022 (black line – median; dark shading – 50% credible interval; light shading – 95% credible interval) is shown relative to the retrospective models. Colored lines correspond to the last year of index data and the colored point indicates the estimate in the last year of the retrospective peel.

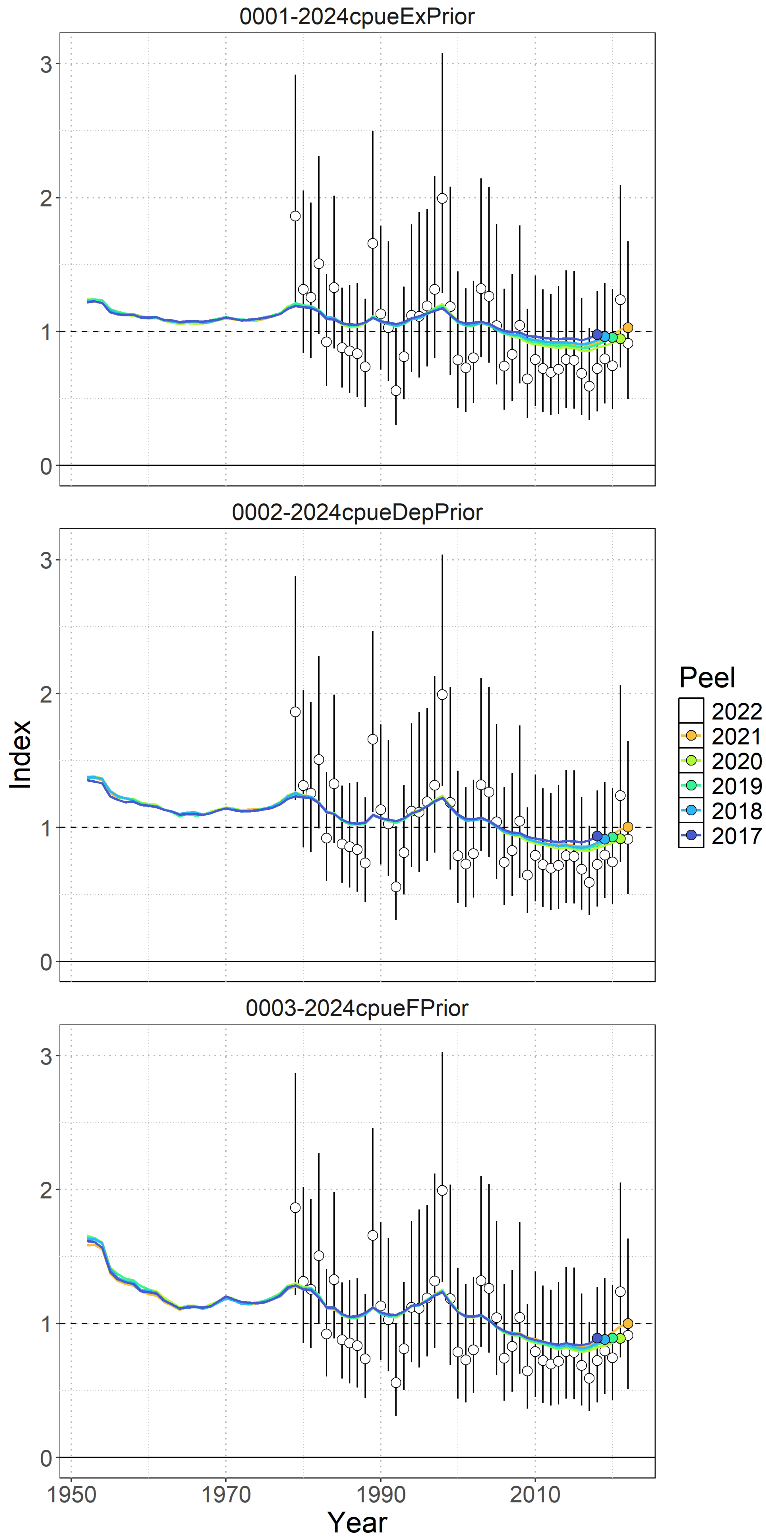

Figure 14: Standardized indices of relative abundance used in the Bayesian Surplus Production Model. Open circles show observed values (standardized to mean of 1; black horizontal line) and the vertical bars indicate the observation error (95% confidence interval). One year ahead 'model-free' hindcast predictions are shown by the colored lines, where the color indicates the last year of index data seen by the model. The predicted value is shown one year-ahead with the colored point.

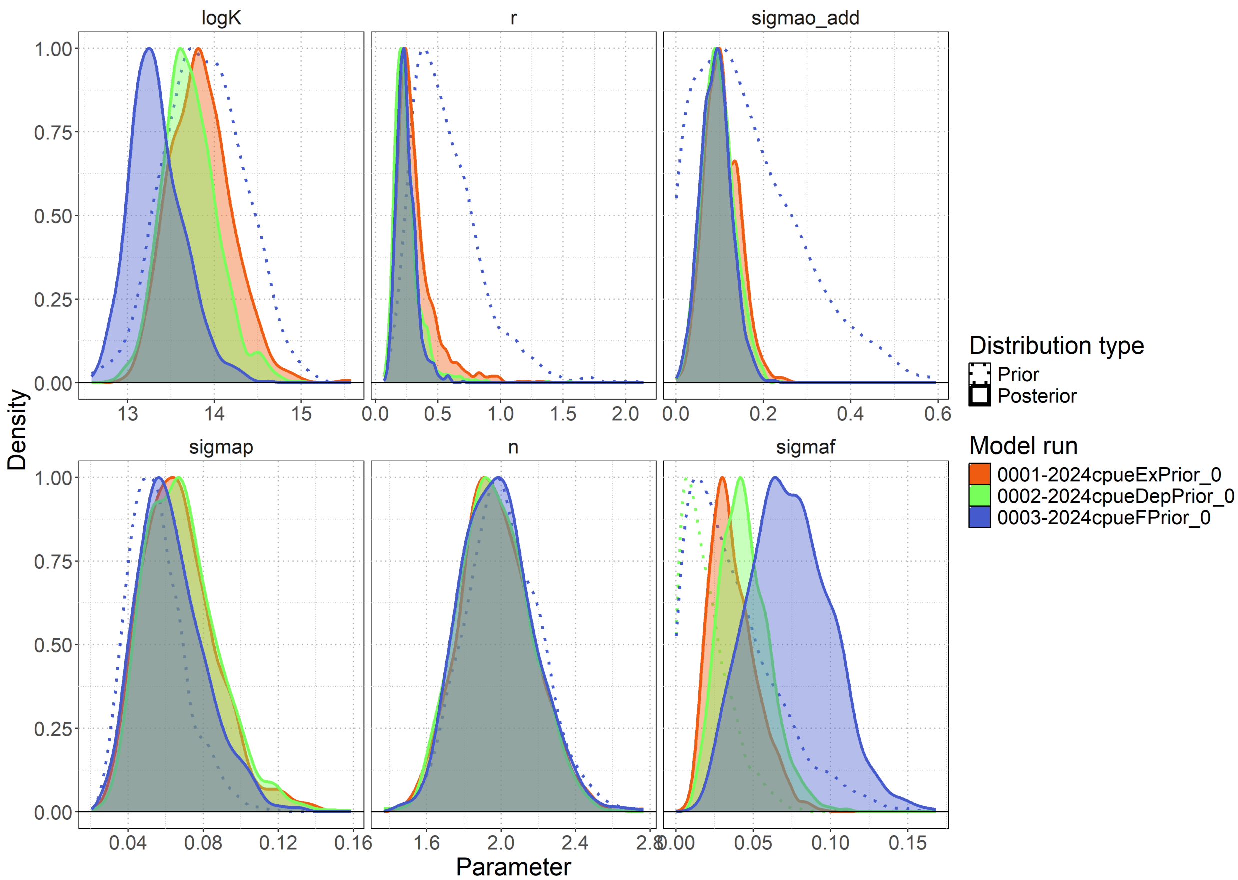

Figure 15: Posterior parameter distributions (solid lines) for leading parameters (*log(K)*, R_{Max}, σ_P, shape (n), σ_F, $\sigma_{O,add}$, and $\sigma_{O,sc}$), relative to their assumed prior distributions (dotted lines) for all models.

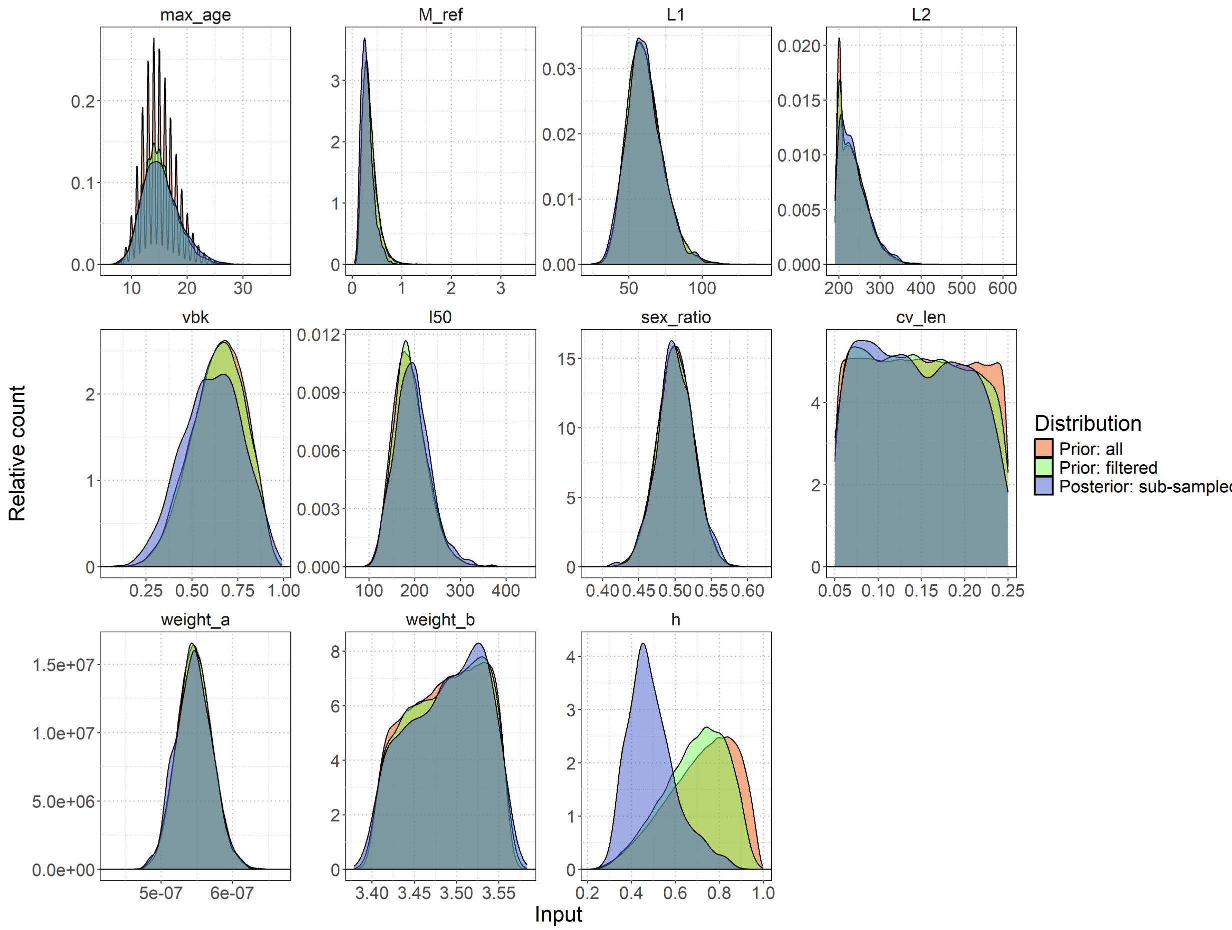

Figure 16: Distributions of biological parameters (maximum age *max_age*, natural mortality reference *M_ref*, length at birth *L1*, asymptotic length *L2*, von Bertalanffy growth coefficient *vbk*, length at 50% maturity *l50*, sex ratio at birth *sex_ratio*, coefficient of variation in length *cv_len*, weight-length relationship parameters *weight_a* and *weight_b*, and steepness *h*) used in the rmax sub-sampling process. Original biological prior distribution (orange shading), distribution based on extreme filtering criteria following the prior pushforward analysis (green shading), and final sub-sampled distribution selected to match the posterior R_{Max} distribution (blue shading).

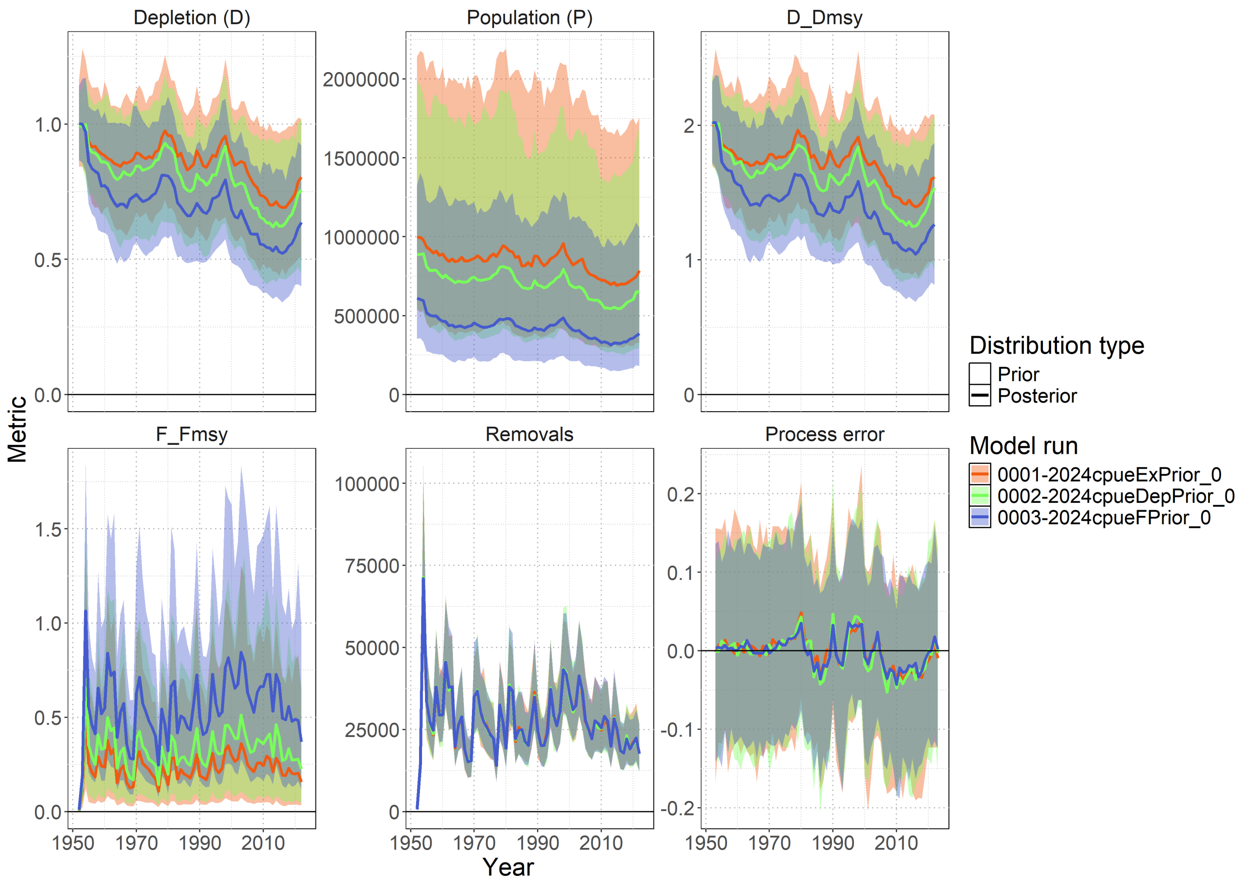

Figure 17: Posterior time series estimates (1952-2022) for key population metrics across all three BSPM models. Solid lines show median estimates with 95% credible intervals (shading). Metrics include depletion (D), population (P), relative depletion (D/D_{MSY}), relative fishing mortality (F/F_{MSY}), removals, and process error.

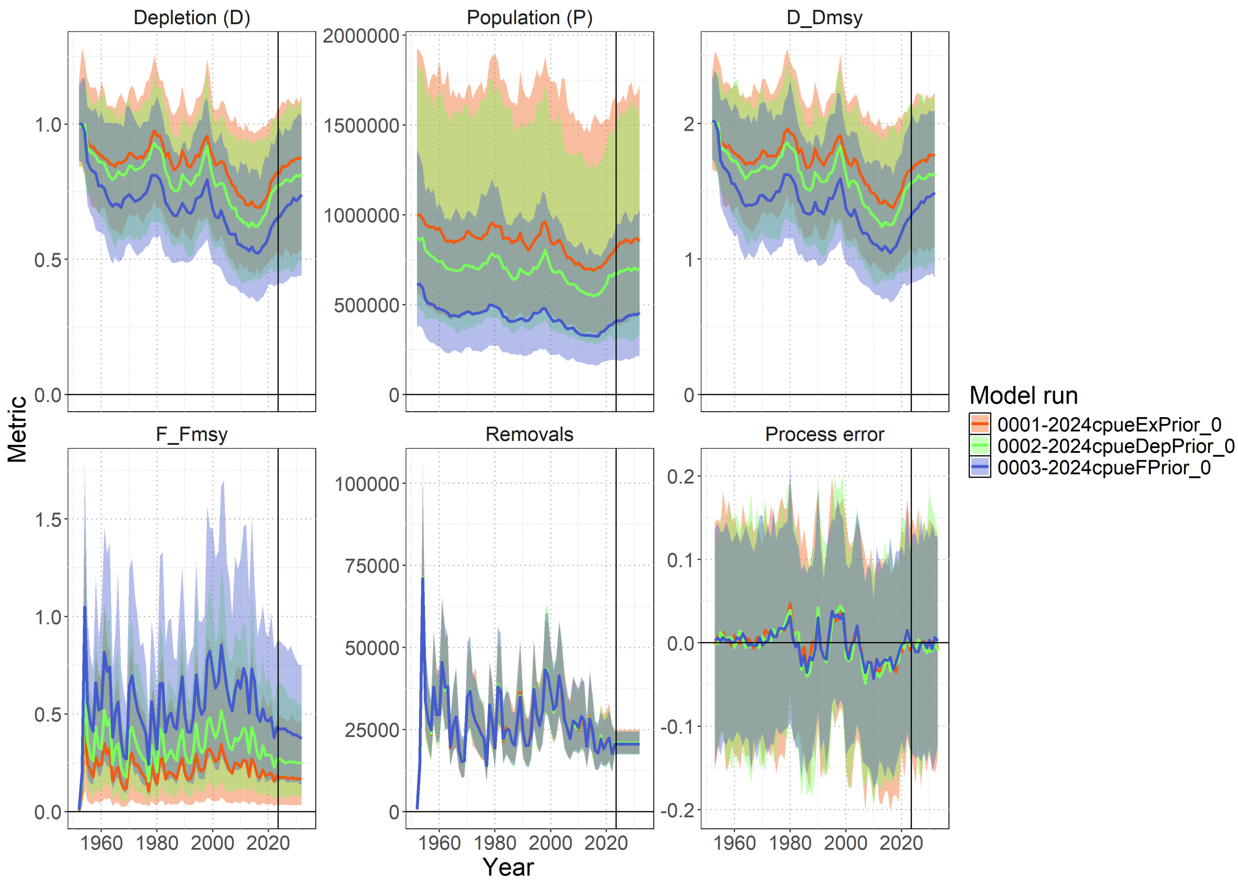

Figure 18: Ten-year stochastic projections (2023-2032) under status quo catch scenario. Colored lines show median projections with 95% credible intervals for all three models. Projections assume average 2018-2022 catch levels with resampled process error. Black vertical line indicates start of projection period. Metrics include depletion (D), population (P), relative depletion (D/D_{MSY}), relative fishing mortality (F/F_{MSY}), removals, and process error.

11 Technical Annex

Additional technical detail of population dynamics components used to solve the Euler-Lotka equation for R_{Max}.

11.1 Equilibrium Age-Structured Population Dynamics

11.1.1 Survival

Survival schedule was calculated assuming constant natural mortality:

$$l_a = \begin{cases} 1 & \text{if } a = 1 \\ l_{a-1} \times \exp(-M_{ref}) & \text{if } 1 < a < A_{Max} \\ \frac{l_{A_{Max}-1}}{1-\exp(-M_{ref})} & \text{if } a = A_{Max} \text{ (plus group)} \end{cases}$$

11.1.2 Fecundity

Reproductive output incorporated biological constraints:

$$f_a = \frac{\psi_a \times W_a \times s}{\rho}$$

where: - ψ_a = proportion mature at age a - W_a = weight at age a (proxy for reproductive capacity) - s = sex ratio (proportion female) - ρ = reproductive cycle (years between spawning events)

Maturity-at-age was derived from length-based logistic maturity:

$$\psi_L = \frac{\exp(a_{mat} + b_{mat} \times L)}{1 + \exp(a_{mat} + b_{mat} \times L)}$$

where $b_{mat} = -a_{mat}/L_{50}$, then integrated over the length-at-age distribution to obtain ψ_a.

Length-at-age followed the Francis parameterization:

$$L_a = L_1 + (L_2 - L_1) \times \frac{1 - \exp(-k(a - age_1))}{1 - \exp(-k(age_2 - age_1))}$$

with length variability modeled using probability density functions linking length bins to ages.

Weight-at-age used the allometric relationship:

$$W_a = a_w \times L_a^{b_w}$$

11.1.3 Stock-Recruitment Relationship

The slope at the origin of the stock-recruitment curve was:

$$\alpha = \frac{4h}{(1-h)\phi_0}$$

The calculation incorporated steepness (h) through the Beverton-Holt stock-recruitment relationship. Spawners-per-recruit in the unfished state was:

$$\phi_0 = \sum_{a=1}^{A_{Max}} l_a f_a$$

This α parameter links the intrinsic rate of increase to recruitment compensation, ensuring that R_{Max} estimates reflect realistic population productivity under the assumed stock-recruitment dynamics.

Successful simulations (those yielding $R_{Max} > 0$ and $R_{Max} < 1.5$) were filtered to create a plausible prior distribution (Figure 3). The resulting distribution was fitted to a lognormal prior for use in the BSPM (Figure 4).